

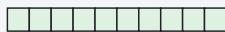
Heterogeneous Data Movement in Persistent RSNN Execution

GLOBAL MEMORY

Static contiguous ownership



V



PSC / input_current

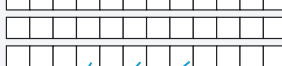


indptr

indices

weights

Global connectivity (CSR)



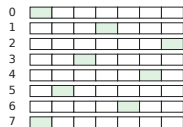
ON-CHIP PERSISTENT CTA

UPDATE — locality-oriented

Static ownership

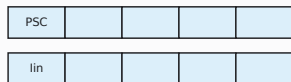
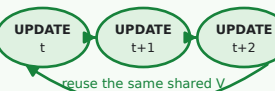
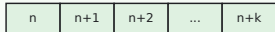
B=1, neurons/CTA <= 256

8 warp lanes



Conditional shared V

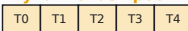
load once before timestep loop



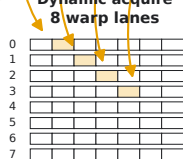
global load/store every timestep

PROP — balance-oriented

Dynamic task pool



Dynamic acquire



Grouped edge access

warp processor
prefix -> logical edges

Edge contributions (post IDs)



Per-warp shared hash



probe fallback

Unique post flush

Global PSC



Temporal reuse across timesteps

Request aggregation within a timestep